```
import tensorflow as tf
```

```
scalar = tf.constant(10)
print("Scalar:")
print(scalar)
```

```
Scalar:
tf.Tensor(10, shape=(), dtype=int32)
```

```
vector = tf.constant([1, 2, 3, 4, 5])
print("Vector:")
print(vector)
```

```
Vector:
tf.Tensor([1 2 3 4 5], shape=(5,), dtype=int32)
```

```
matrix = tf.constant([[1, 2],
                      [3, 4]])
print("Matrix:")
print(matrix)
```

```
Matrix:
tf.Tensor(
[[1 2]
 [3 4]], shape=(2, 2), dtype=int32)
```

```
tensor3d = tf.constant([
    [[1, 2], [3, 4]],
    [[5, 6], [7, 8]]
])
print("3-D Tensor:")
print(tensor3d)
```

```
3-D Tensor:
tf.Tensor(
[[[1 2]
  [3 4]]

 [[5 6]
  [7 8]]], shape=(2, 2, 2), dtype=int32)
```

```
# Create a tensor
tensor = tf.constant([[10, 20, 30],
                      [40, 50, 60]])

# Display tensor properties
print("Tensor:")
print(tensor)

print("\nShape:", tensor.shape)
print("Data Type:", tensor.dtype)
print("Dimensions:", tensor.ndim)
print("Number of Elements:", tf.size(tensor).numpy())
```

```
Tensor:
tf.Tensor(
[[10 20 30]
 [40 50 60]], shape=(2, 3), dtype=int32)

Shape: (2, 3)
Data Type: <dtype: 'int32'>
Dimensions: 2
Number of Elements: 6
```

```
# Create two tensors
A = tf.constant([10, 20, 30])
B = tf.constant([1, 2, 3])

# Arithmetic Operations
print("Tensor A:", A)
print("Tensor B:", B)

print("\nAddition:")
print(tf.add(A, B))

print("\nSubtraction:")
print(tf.subtract(A, B))
```

```
print("\nMultiplication:")
print(tf.multiply(A, B))

print("\nDivision:")
print(tf.divide(A, B))
```

```
Tensor A: tf.Tensor([10 20 30], shape=(3,), dtype=int32)
Tensor B: tf.Tensor([1 2 3], shape=(3,), dtype=int32)

Addition:
tf.Tensor([11 22 33], shape=(3,), dtype=int32)

Subtraction:
tf.Tensor([ 9 18 27], shape=(3,), dtype=int32)

Multiplication:
tf.Tensor([10 40 90], shape=(3,), dtype=int32)

Division:
tf.Tensor([10. 10. 10.], shape=(3,), dtype=float64)
```

```
# Create two matrices
A = tf.constant([[1, 2],
                 [3, 4]])

B = tf.constant([[5, 6],
                 [7, 8]])

print("Matrix A:")
print(A)

print("\nMatrix B:")
print(B)

# Matrix Addition
print("\nMatrix Addition:")
print(tf.add(A, B))

# Matrix Multiplication
print("\nMatrix Multiplication:")
print(tf.matmul(A, B))

# Transpose of Matrix A
print("\nTranspose of Matrix A:")
print(tf.transpose(A))

# Transpose of Matrix B
print("\nTranspose of Matrix B:")
print(tf.transpose(B))
```

```
Matrix A:
tf.Tensor(
[[1 2]
 [3 4]], shape=(2, 2), dtype=int32)

Matrix B:
tf.Tensor(
[[5 6]
 [7 8]], shape=(2, 2), dtype=int32)

Matrix Addition:
tf.Tensor(
[[ 6  8]
 [10 12]], shape=(2, 2), dtype=int32)

Matrix Multiplication:
tf.Tensor(
[[19 22]
 [43 50]], shape=(2, 2), dtype=int32)

Transpose of Matrix A:
tf.Tensor(
[[1 3]
 [2 4]], shape=(2, 2), dtype=int32)

Transpose of Matrix B:
tf.Tensor(
[[5 7]
 [6 8]], shape=(2, 2), dtype=int32)
```

```
# Create a matrix
A = tf.constant([[1, 2, 3],
                 [4, 5, 6]])

print("Matrix A:")
```

```
print(A)

# Sum of all elements
print("\nSum of all elements:")
print(tf.reduce_sum(A))

# Sum of each column
print("\nColumn-wise Sum:")
print(tf.reduce_sum(A, axis=0))

# Sum of each row
print("\nRow-wise Sum:")
print(tf.reduce_sum(A, axis=1))

# Mean of all elements
print("\nMean of all elements:")
print(tf.reduce_mean(tf.cast(A, tf.float32)))

# Mean of each column
print("\nColumn-wise Mean:")
print(tf.reduce_mean(tf.cast(A, tf.float32), axis=0))

# Mean of each row
print("\nRow-wise Mean:")
print(tf.reduce_mean(tf.cast(A, tf.float32), axis=1))
```

```
# Create a 2-D tensor
tensor = tf.constant([[10, 20, 30],
                      [40, 50, 60],
                      [70, 80, 90]])

print("Original Tensor:")
print(tensor)

# Indexing
print("\nFirst Row:")
print(tensor[0])

print("\nSecond Row:")
print(tensor[1])

print("\nElement at Row 2, Column 3:")
print(tensor[1, 2])

# Slicing
print("\nFirst Two Rows:")
print(tensor[:2])

print("\nSecond Column:")
print(tensor[:, 1])

print("\nLast Two Columns:")
print(tensor[:, 1:])
```

```
Original Tensor:
tf.Tensor(
[[10 20 30]
 [40 50 60]
 [70 80 90]], shape=(3, 3), dtype=int32)

First Row:
tf.Tensor([10 20 30], shape=(3,), dtype=int32)

Second Row:
tf.Tensor([40 50 60], shape=(3,), dtype=int32)

Element at Row 2, Column 3:
tf.Tensor(60, shape=(), dtype=int32)

First Two Rows:
tf.Tensor(
[[10 20 30]
 [40 50 60]], shape=(2, 3), dtype=int32)

Second Column:
tf.Tensor([20 50 80], shape=(3,), dtype=int32)

Last Two Columns:
tf.Tensor(
[[20 30]
 [50 60]
 [80 90]], shape=(3, 2), dtype=int32)
```

```
# Create a TensorFlow Variable
x = tf.Variable([10, 20, 30])

print("Original Variable:")
print(x)

# Update the variable
x.assign([100, 200, 300])

print("\nAfter assign():")
print(x)

# Add a value to each element
x.assign_add([10, 10, 10])

print("\nAfter assign_add():")
print(x)

# Subtract a value from each element
x.assign_sub([5, 5, 5])

print("\nAfter assign_sub():")
print(x)
```

```
Original Variable:
<tf.Variable 'Variable:0' shape=(3,) dtype=int32, numpy=array([10, 20, 30], dtype=int32)>

After assign():
<tf.Variable 'Variable:0' shape=(3,) dtype=int32, numpy=array([100, 200, 300], dtype=int32)>

After assign_add():
<tf.Variable 'Variable:0' shape=(3,) dtype=int32, numpy=array([110, 210, 310], dtype=int32)>

After assign_sub():
<tf.Variable 'Variable:0' shape=(3,) dtype=int32, numpy=array([105, 205, 305], dtype=int32)>
```

Start coding or generate with AI.